



Technical Service Bulletin: tsb101150

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Fuel Inlet Check Valve and 10 Micron Fuel Filter

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This service/parts topic provides information on the improved fuel filtration capability and updates procedure 018-024.

Improved fuel filtration capability has been introduced for the QST30 industrial engines. Industrial engines are now using Fleetguard FS1006 filters, Cummins Part Number 3089916. The FS1006 offers 10 micron [.0004 in] filtration and water separation capability, and will mount to the existing filter head on all QST30 engines. The FS1006 is a direct replacement for the FF202 filters used previously.

When the water drain valve on the bottom of the filter is opened, a drain-back path exists for fuel to return to the tank when the tank fuel level is below filter level. This condition introduces air into the supply lines, and could result in hard starting. Drainback could also occur if there was any air leaks into the fuel system.

A check valve has been added at the fuel inlet manifold, to reduce the possibility of drain-back. The check valve includes a 300 micron filter screen inside the fitting. The filter screen can be removed for cleaning by isolating the fuel tank, disconnecting the fuel supply hose, and removing the fitting from the check valve. The check valve remains in the fuel manifold to reduce the possibility of fuel from draining out of the engine. The screen is then removed from the fitting for cleaning.

In the event of a low power complaint or the presence of Fault Code 423 or 423R, the filter screen should be inspected for debris.

A 27.6 kPa [4 psi] banjo check valve relieves pressure from the supply to the drain, to reduce the possibility of fuel filter seal damage due to fuel expansion after shutdown. This check valve is mounted on top of the fuel manifold on the supply side.

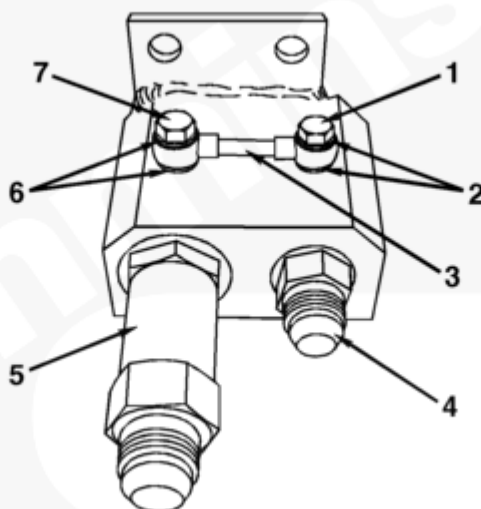
Engine serial number first for Euclid engines with the fuel inlet check valve and FS1006 fuel filters is 37198242, built on 16 April 2001. Engine serial number first for Komatsu engines is 37202119, built on 28 January 2002. Engine serial number first for all other OEM'S IS 37200921, built on 15 October 2001.

A kit is available to upfit QST30 Industrial engines built prior to the engine serial number first listed above. The kit part number is 4089113. In addition to the kit, it is also necessary to order the appropriate fuel manifold for the engine. The fuel manifold required is based on the FT (Fuel Tubing) option the engine uses. The fuel tubing option content is:

Option	Manifold
FT5709	4067988
All others	4067987

Note : The fuel manifold is not included with the upfit kit and must be ordered separately.

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Fuel Filter Upfit Kit Part Numbers

Reference Number	Part Number	Quantity	Description	Torque
Not shown	3089916	2	Filter, fuel (FS1006)	
1	3094188	1	Bolt, banjo	7 N•m [62 in-lb]
2	3918188	2	Washer, sealing (M8)	
3	4067979	1	Tube, fuel by-pass	
4	4067813	1	Fitting, fuel drain	80 N•m [59 ft-lb]
5	4067950	1	Fitting, fuel supply	80 N•m [59 ft-lb]
6	2918190	2	Washer, sealing (M10)	

Fuel Filter Upfit Kit Part Numbers				
Reference Number	Part Number	Quantity	Description	Torque
7	4067977	1	Check valve, banjo	11 N•m [97 in-lb]
Note : This is a product improvement and is notsubject to campaign.				